The documentation and process conversion measures necessary to comply with this document shall be completed by 31 June 2005.

INCH-POUND

MIL-PRF-19500/496C
31 March 2005
SUPERSEDING
MIL-PRF-19500/496B
15 September 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DUAL TRANSISTOR, PNP, SILICON, UNITIZED,
TYPES 2N5795, 2N5796, AND 2N5796U, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for two electrically isolated, unmatched PNP silicon transistors as one dual unit for HI-speed saturated switching applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-78) and figure 2 (U).

* 1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| $P_T$ (1, 2) $T_A$ = +25°C | | $I_C$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $T_J$ and $T_{STG}$ | $R_{\theta JA}$ | $R_{\theta JA}$ |
|---|---|---|---|---|---|---|---|---|
| One section | Total device | | | | | | One section | Total device |
| W | W | mA dc | V dc | V dc | V dc | °C | °C/W | °C/W |
| 0.5 | 0.6 | 600 | 60 | 60 | 5.0 | -65 to +175 | 350 | 290 |

(1) For $T_A \geq 25$°C, derate linearly 2.86 mW/°C one section, 3.43 mW/°C total. For 2N5795
(2) For 2N5795, 2N5796, and 2N5796U devices.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://www.dodssp.daps.mil/.



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* 1.4 Primary electrical characteristics. $T_A$ = +25°C, unless otherwise specified.

| | $C_{obo}$ $V_{CB}$ = 10 V dc $I_E$ = 0 100 kHz ≤ f ≤ 1 MHz | $\|h_{fe}\|$ $V_{CE}$ = 20 V dc $I_C$ = 20 mA dc f = 100 MHz | Switching | |
|---|---|---|---|---|
| | | | $t_{on}$ | $t_{off}$ |
| | pF | | ns | ns |
| Minimum | | 2.0 | | |
| Maximum | 8.0 | 10.0 | 50 | 140 |

| Limits | $h_{FE1}$ $V_{CE}$ = 10 V dc $I_C$ = 100 µA dc | | $h_{FE4}$ (1) $V_{CE}$ = 10 V dc $I_C$ = 150 mA dc | | $V_{CE(SAT)1}$ (1) $I_C$ = 150 mA dc $I_B$ = 15 mA dc | $V_{CE(SAT)2}$ (1) $I_C$ = 500 ma dc $I_B$ = 50 mA dc | $V_{BE(SAT)1}$ (1) $I_C$ = 150 mA dc $I_B$ = 15 mA dc |
|---|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | V dc | V dc | V dc |
| 2N5795 | 40 | | 40 | 150 | 0.4 | 1.6 | 1.3 |
| 2N5796, U | 75 | | 100 | 300 | 0.4 | 1.6 | 1.3 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

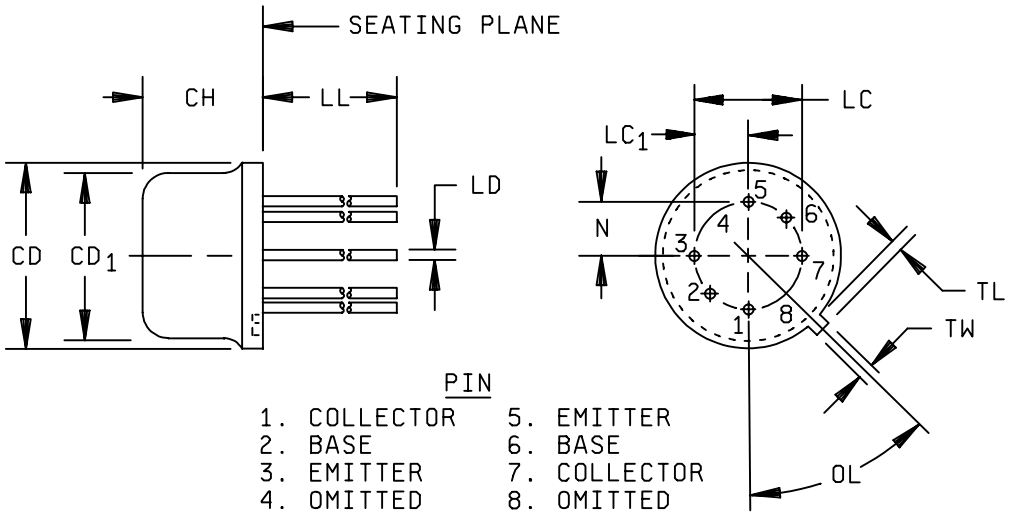
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://www.dodssp.daps.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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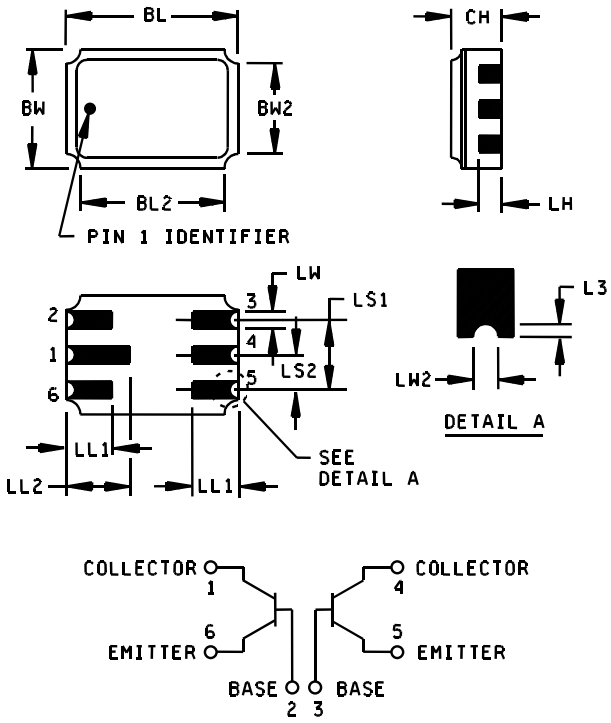

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .335 | .370 | 8.51 | 9.40 | |
| $CD_1$ | .305 | .335 | .335 | 8.51 | |
| CH | .150 | .185 | 3.81 | 4.70 | |
| LD | .016 | .021 | 0.41 | 0.53 | |
| LC | 200 BSC | | 5.08 BSC | | 4 |
| $LC_1$ | 100 BSC | | 2.54 BSC | | 4 |
| LL | .500 | | 12.70 | | |
| TW | .028 | .034 | 0.71 | 0.86 | |
| TL | .029 | .045 | 0.74 | 1.14 | 3 |
| OL | 45° BSC | | 45° BSC | | 6 |
| N | .100 BSC | | 2.54 BSC | | |

NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Measured from maximum diameter of the product.
4. Leads having maximum diameter .019 inch (0.483 mm) measured in gaging plan .054 inch (1.37 mm) + .001 inch (0.025 mm) - .000 inch (0.000 mm) below the seating plane of the product shall be within .007 inch (.178 mm) of their true position relative to a maximum width tab.
5. The product may be measured by direct methods or by gauge.
6. Tab centerline.
7. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions for 2N5795 and 2N5796 (TO-78).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .240 | .250 | 6.10 | 6.35 |
| $BL_2$ | | .250 | | 6.35 |
| BW | .165 | .175 | 4.19 | 4.45 |
| $BW_2$ | | .175 | | 4.45 |
| CH | .058 | .100 | 1.47 | 2.54 |
| $L_3$ | .003 | .007 | 0.08 | 0.18 |
| LH | .026 | .039 | 0.66 | 0.99 |

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| $LL_1$ | .060 | .070 | 1.52 | 1.78 |
| $LL_2$ | .082 | .098 | 2.08 | 2.49 |
| $LS_1$ | .095 | .105 | 2.41 | 2.67 |
| $LS_2$ | .045 | .055 | 1.14 | 1.40 |
| LW | .022 | .028 | 0.56 | 0.71 |
| $LW_2$ | .006 | .022 | 0.15 | 0.56 |
| | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension CH controls the overall package thickness.
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions $LW_2$ minimum and $L_3$ minimum and the appropriate castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension $LW_2$ maximum and $L_3$ maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.
6. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 2. Physical dimensions, 2N5796U.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.4).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (similar to T0-99), and 2 (surface mount) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

* 4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| 3c | Thermal impedance (see 4.3.2) |
| 7 | Optional |
| 9 | Not applicable |
| 10 | 48 hours minimum. |
| 11 | $I_{CBO2}$ and $h_{FE4}$ |
| 12 | (See 4.3.1) |
| 13 | Subgroup 2 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc; whichever is greater.<br>$\Delta h_{FE4}$ = ± 15 percent of initial value. |
| 14 | Required |

* 4.3.1 Burn-in conditions. Burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc; power shall be applied to achieve $T_J$ = +135°C minimum using $P_T$ = 300 mW each section (600 mW total device). $T_A$ = ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, TJ, and mounting conditions) may be used. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750.

a. $I_M$ measurement current----------5 mA.

b. $I_H$ forward heating current ----------200 mA (min).

c. $t_H$ heating time ----------25 - 30 ms.

d. $t_{md}$ measurement delay time --------60 µs max.

e. $V_{CE}$ collector-emitter voltage --------10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I subgroup 1 and 2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500. (End-point electrical measurements shall be in accordance with the applicable tests of table I, subgroup 2 and 4.5.2 herein

* 4.4.2.1 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and /or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +175°C. n = 22, c = 0. |

* 4.4.2.2 Group B sample selection. Samples selected for group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and in 4.4.3.1 herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable tests of table I, subgroup 2 and 4.5.2 herein).

* 4.4.3.1 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E, not applicable for “U” designated devices. |
| C5 | 3131 | $R_{\theta JA}$, see 1.3. |
| C6 | | Not applicable. |

4.4.3.2 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.



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* 4.4.3.3 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit |
|---|---|---|---|---|---|
| | | Method | Conditions | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 50 V dc | $\Delta I_{CB02}$ (1) | 100 percent of initial value or ± 5 nA dc, whichever is greater. |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed see 4.5.1 | $\Delta h_{FE4}$ (1) | ±25 percent change from initial reading. |

(1) Devices which exceed the table I limits for this test shall not be accepted.



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TABLE I. Group A inspection

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical Examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/, 4/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temprature cycling 3/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Heremetic seal<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements | | Table I, subgroup 2 | | | | |
| Bond strength 3/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hours or $T_A$ = +300°C at t = 2 hours n = 11 wires, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.2. | $Z_{\theta JX}$ | | | °C/W |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CBO}$ = 60 V dc | $I_{CBO1}$ | | 10 | µA dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 60 | | V dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO1}$ | | 10 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 50 V dc | $I_{CBO2}$ | | 10 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 3 V dc | $I_{EBO2}$ | | 100 | nA dc |
| Forward-current transfer ratio<br>2N5795<br>2N5796, 2N5796U | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 100 µA dc | $h_{FE1}$ | <br>40<br>75 | | |
| Forward-current transfer ratio<br>2N5795<br>2N5796, 2N5796U | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc | $h_{FE2}$ | <br>40<br>100 | | |
| Forward-current transfer ratio<br>2N5795<br>2N5796, 2N5796U | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 10 mA dc pulsed (see 4.5.1) | $h_{FE3}$ | <br>40<br>100 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued. | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N5795 | | | | 40 | 150 | |
| 2N5796, 2N5796U | | | | 100 | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 300 mA dc; pulsed (see 4.5.1) | $h_{FE5}$ | | | |
| 2N5795 | | | | 20 | | |
| 2N5796, 2N5796U | | | | 50 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 150 mA dc; pulsed (see 4.5.1) | $h_{FE6}$ | | | |
| 2N5795 | | | | 20 | | |
| 2N5796, 2N5796U | | | | 50 | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 150 mA dc; $I_B$ = 15 mA dc pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.4 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 1.6 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 150 mA dc; $I_B$ = 15 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.3 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 2.6 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 50 V dc | $I_{CBO3}$ | | 10 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc | $h_{FE7}$ | | | |
| 2N5795 | | | | 16 | | |
| 2N5796, 2N5796U | | | | 40 | | |
| Subgroup 4 | | | | | | |
| Magnitude of small-signal short- circuit forward current transfer ratio | 3306 | $V_{CE}$ = 20 V dc; $I_C$ = 20 mA dc; f = 100 MHz | $\lvert h_{fe} \rvert$ | 2 | 10 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 8 | pF |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued. | | | | | | |
| Input capacitance (output open- circuited) | 3240 | $V_{EB}$ = 0.5 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 25 | pF |
| Pulse response | 3251 | Test condition A, (see figure 3) | | | | |
| Saturated turn-on time | | $V_{CC}$ = 30 V dc; $I_C$ = 150 mA dc; $I_{B1}$ = 15 mA dc, $V_{BE(OFF)}$ = 0.5 V dc | $t_{on}$ | | 50 | ns |
| Saturated turn-off time | | $V_{CC}$ = 30 V dc; $I_C$ = 150 mA dc; $I_{B1}$ = $I_{B2}$ =15 mA dc | $t_{off}$ | | 140 | ns |
| Subgroup 5 | | | | | | |
| Collector one to collector two leakage current | | $V_{(1C-2C)}$ = ±50 V dc. | $1_{(1C-2C)}$ | | ±1 | NA dc |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Decap internal visual (design verification) | 2075 | n = 1 device, c = 0 | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed table I, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.



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* TABLE II. Group E inspection (all quality levels) - for qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.2 herein. | 45 devices c = 0 |
| Subgroup 2 | | | |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles. | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.2 herein. | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | Each supplier shall submit their qualification lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices c = 0 |
| ESD | 1020 | | |
| Subgroup 8 | | | 45 devices c = 0 |
| Reverse stability | 1033 | Condition B | |



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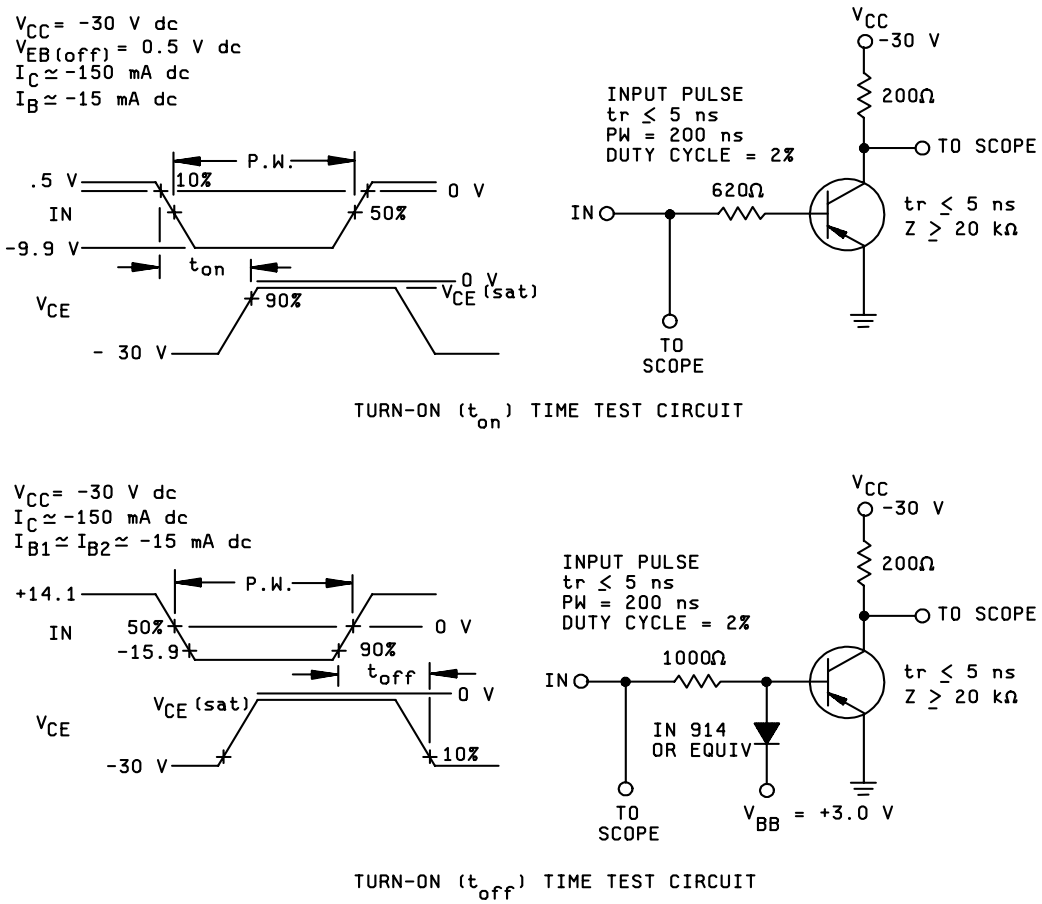



FIGURE 3. Switching time test circuits.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2819)

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS
Air Force - 19

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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